So what do networks do?

Example: Classification

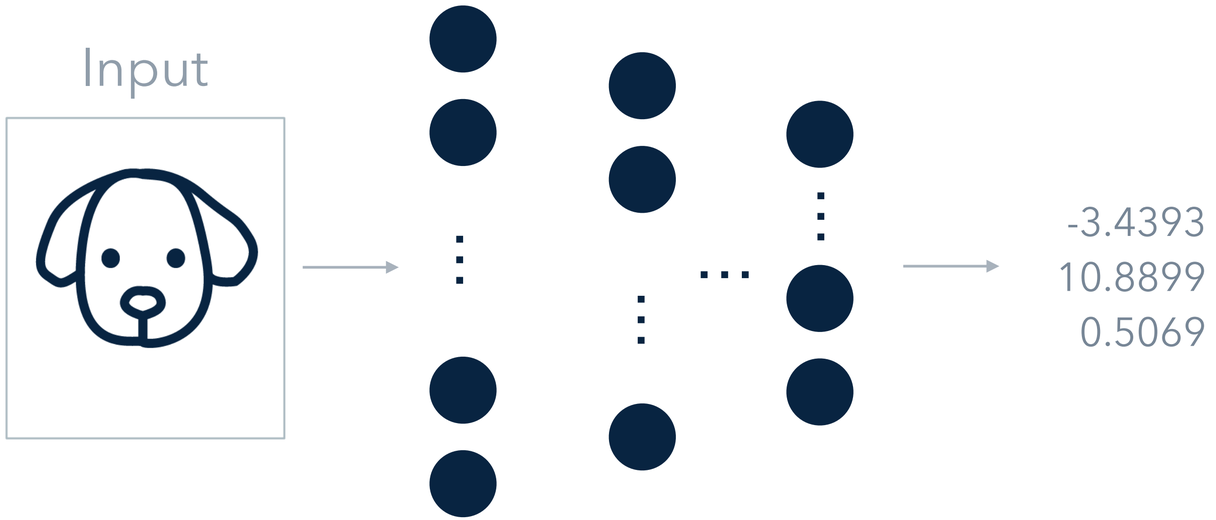


vs

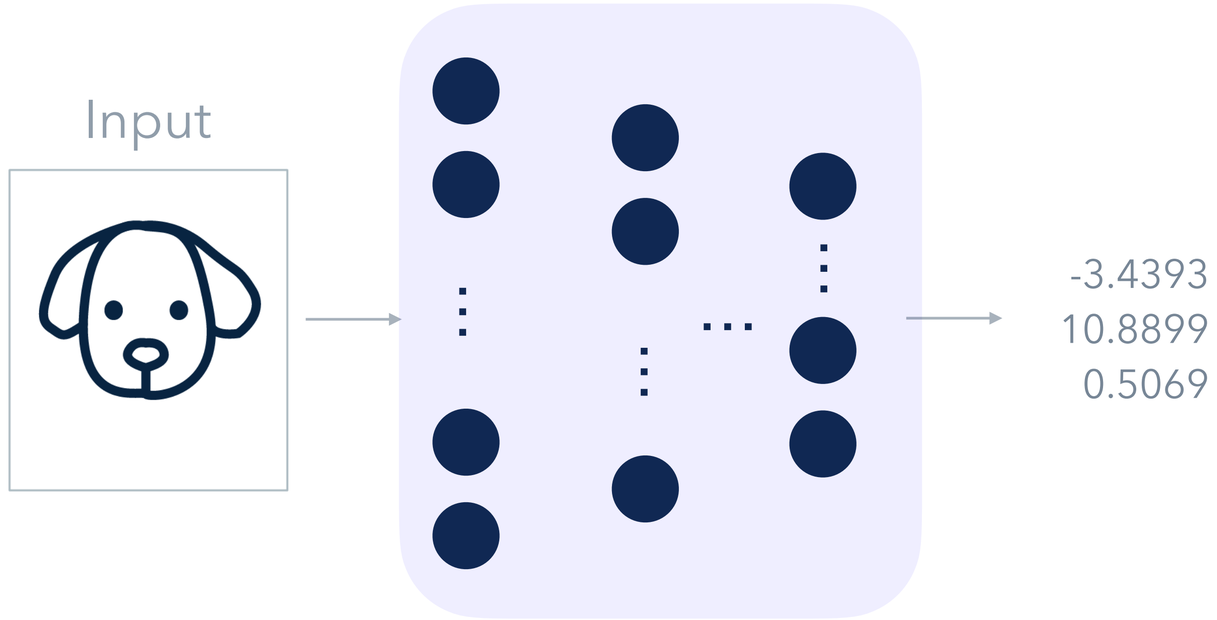


vs

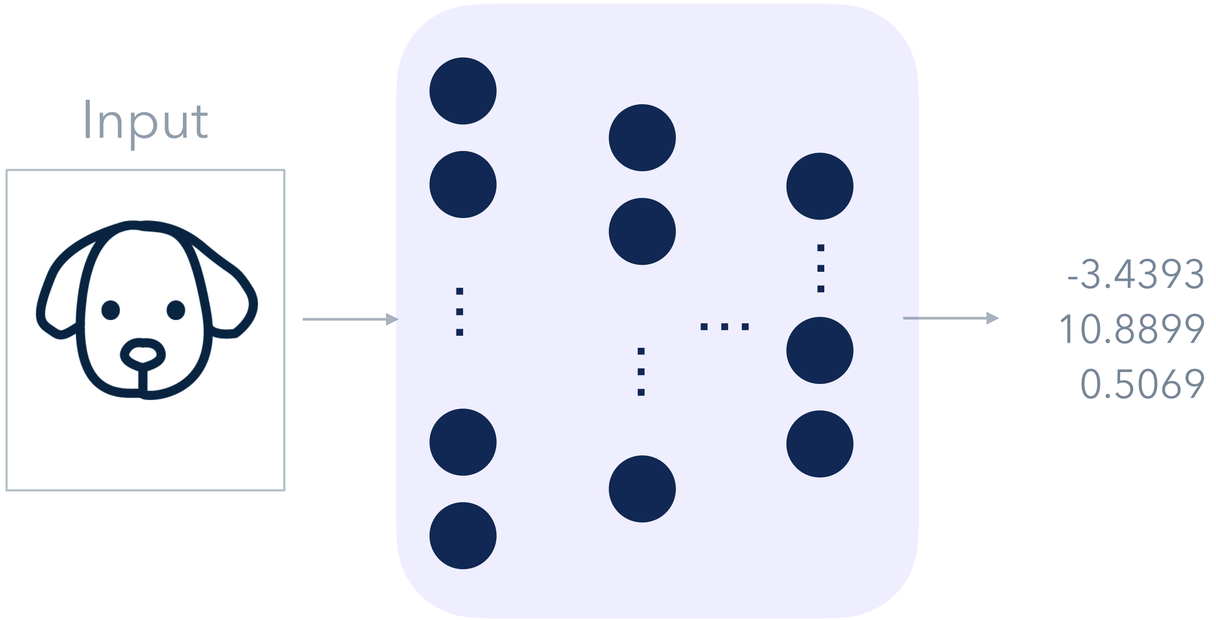



Input
-3.4393
10.8899
0.5069

So far

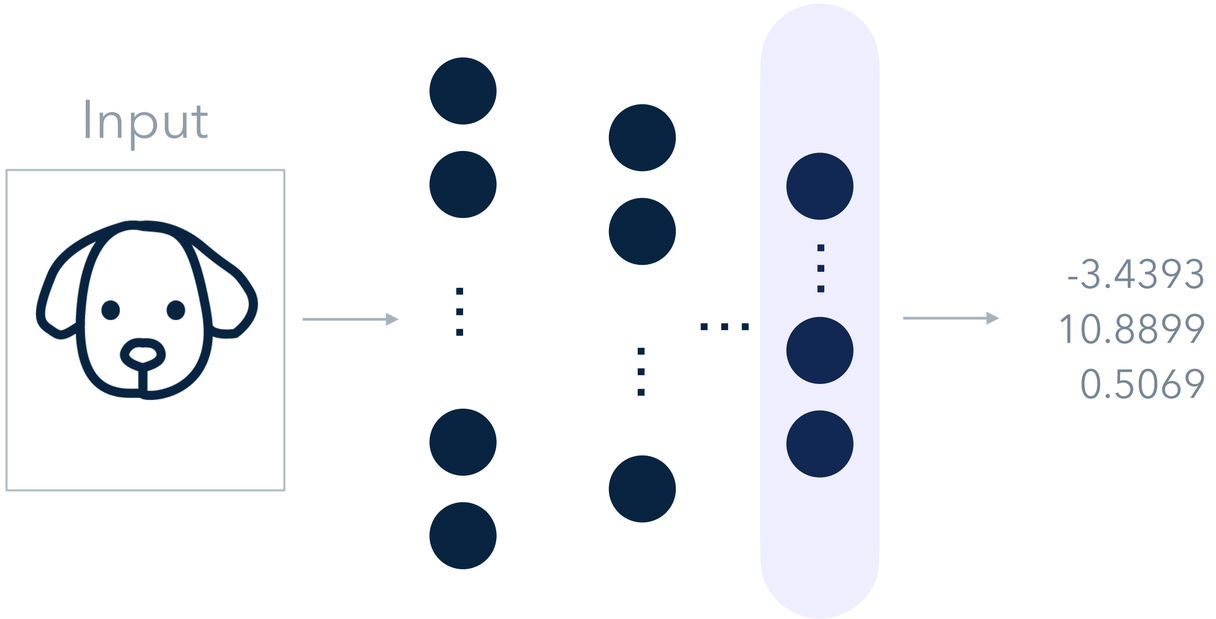


So far

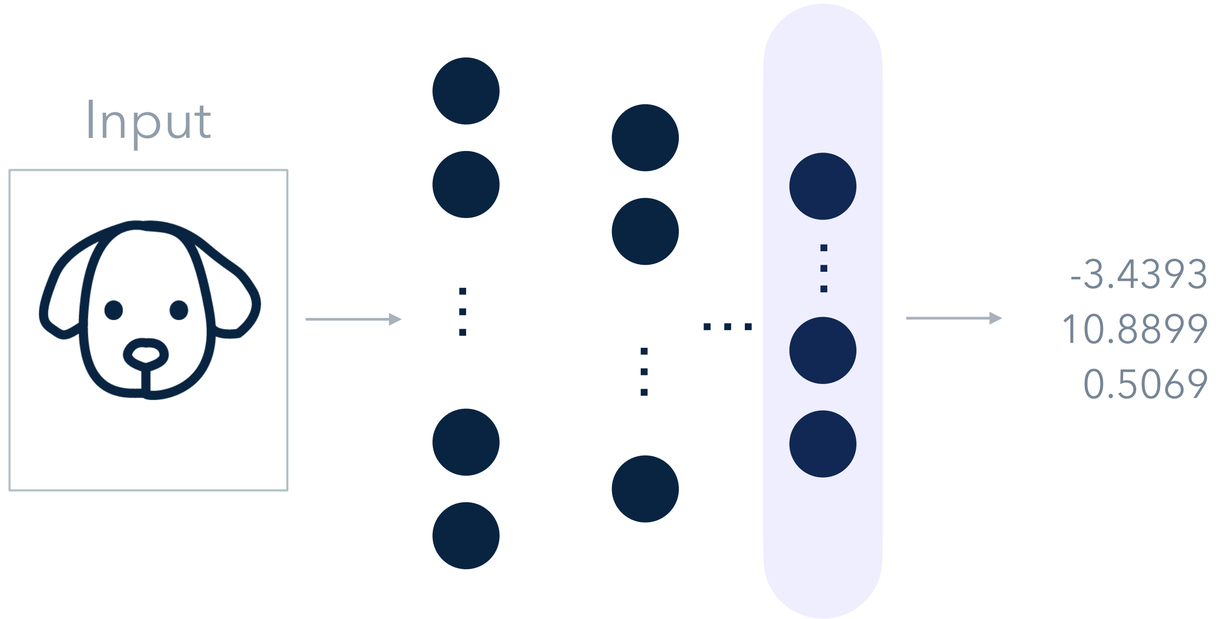


So how does a network solve a classification task?

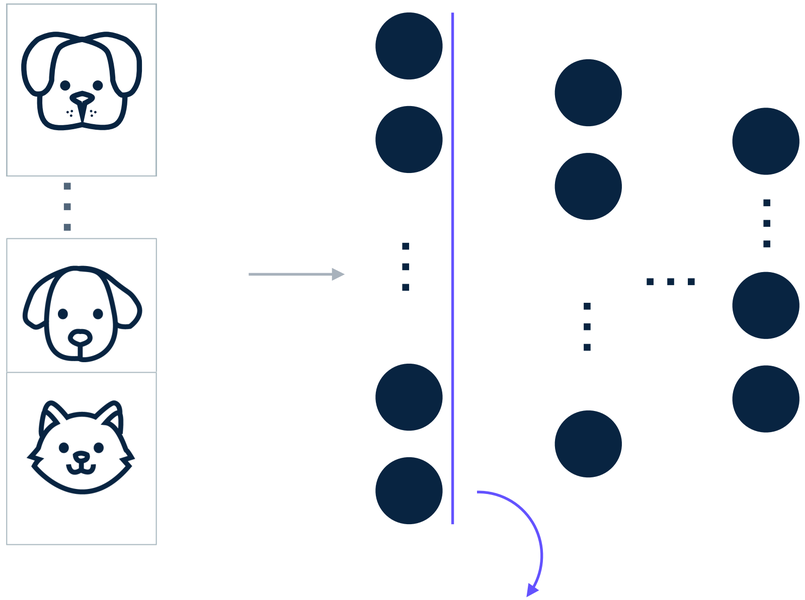
Let’s start from the end



Making data linearly separable

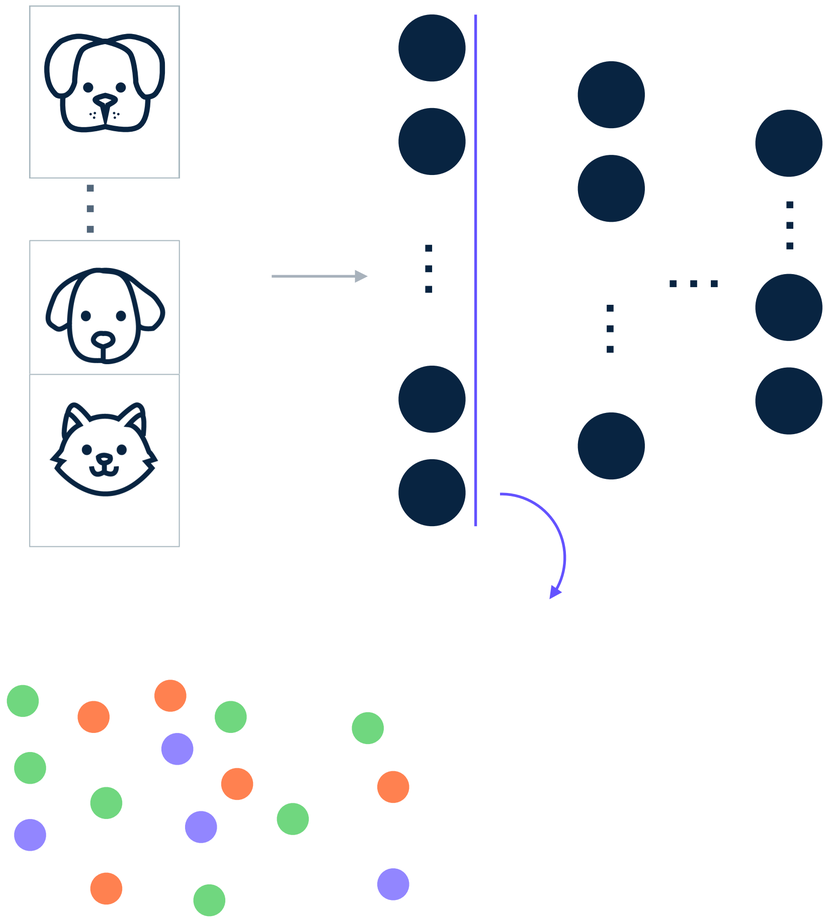


Making data linearly separable

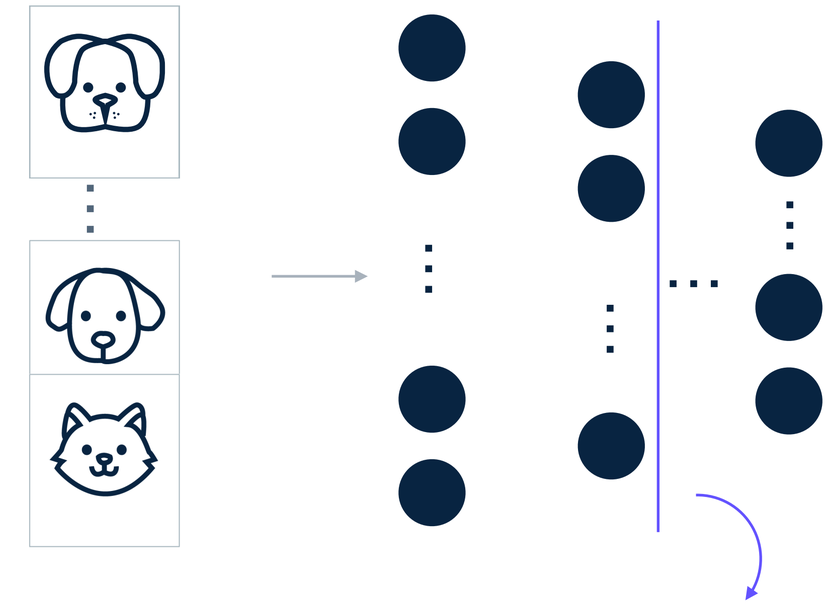


What do we obtain here?

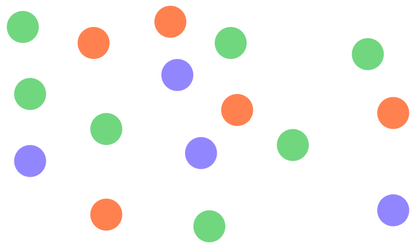
Making data linearly separable



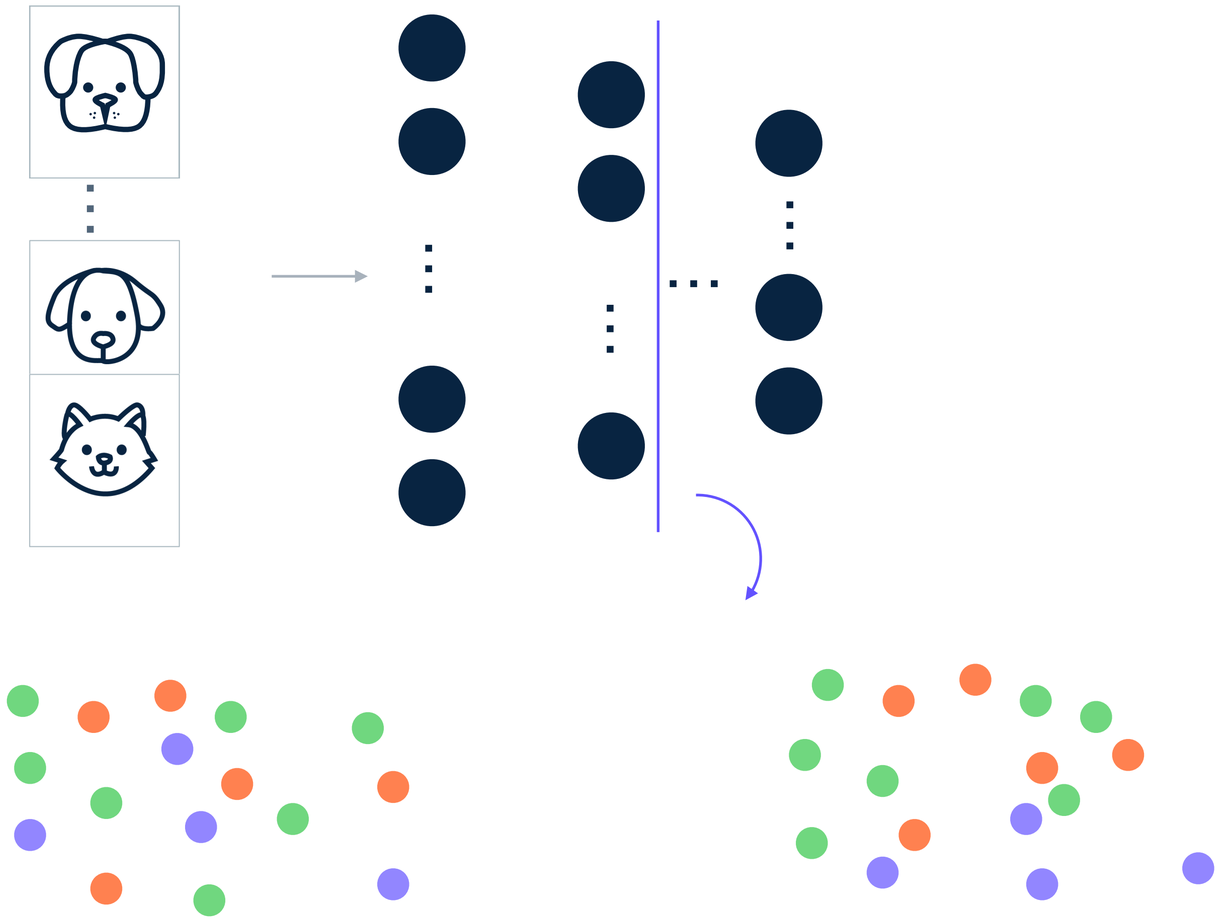
Making data linearly separable



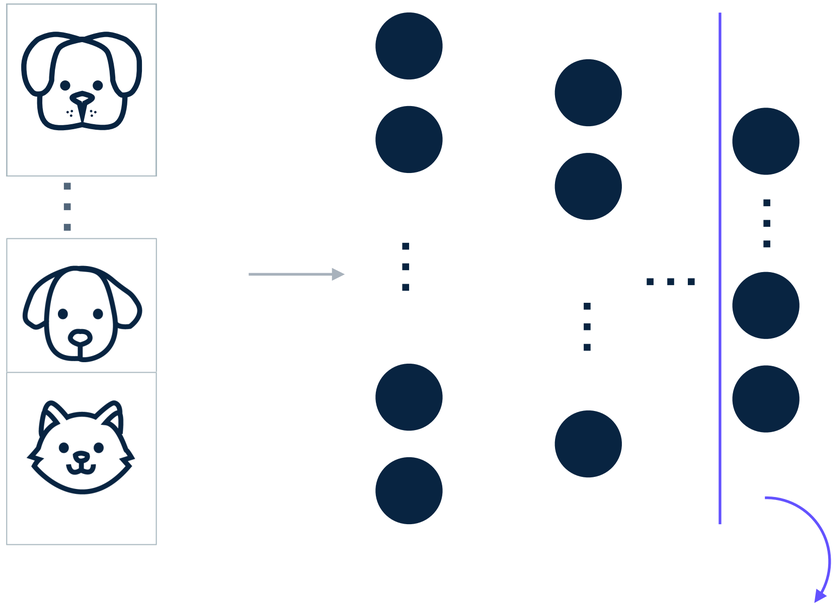
What might we expect to change?



Making data linearly separable

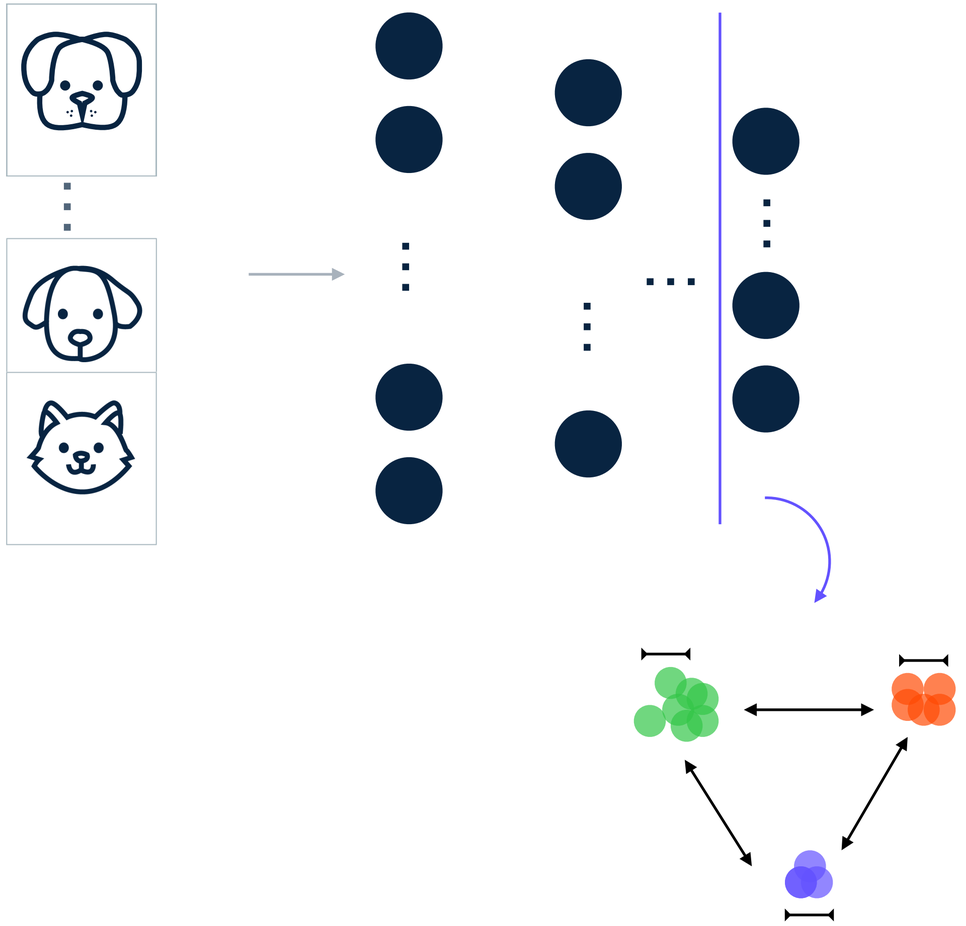


Making data linearly separable

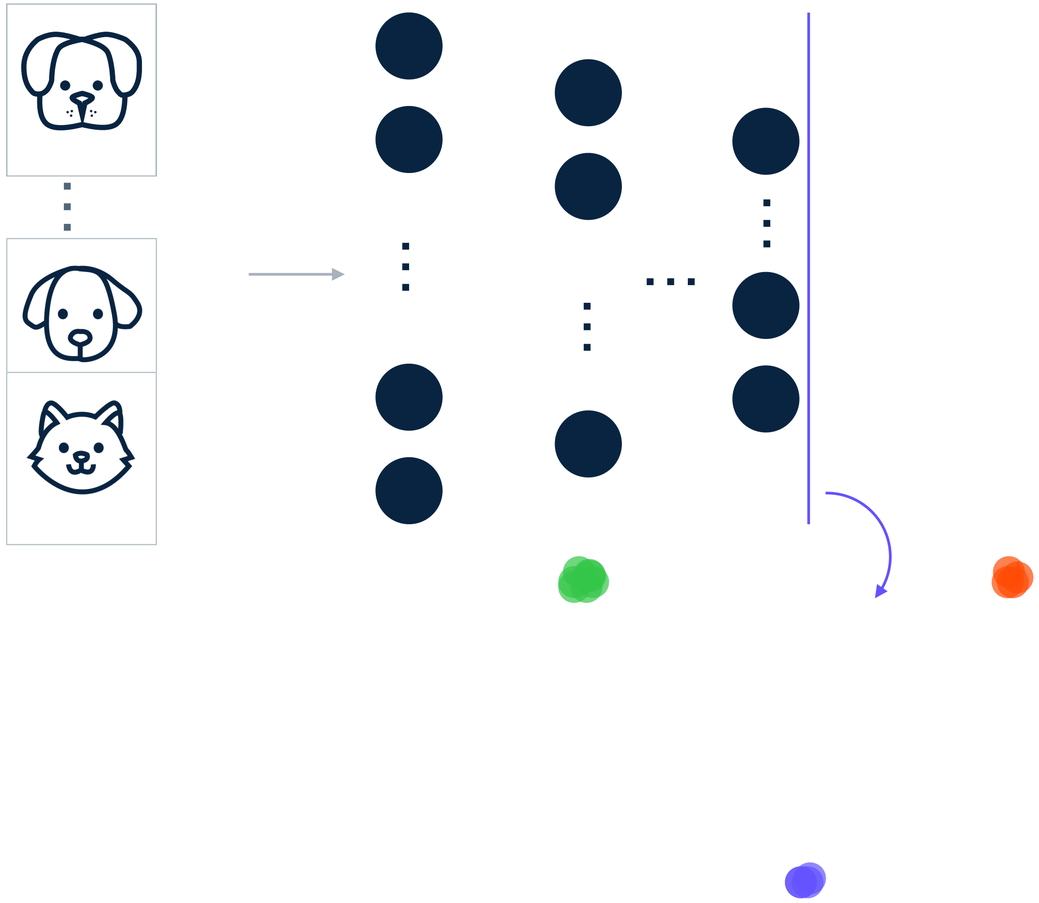


What does CE push representations towards?

Making data linearly separable



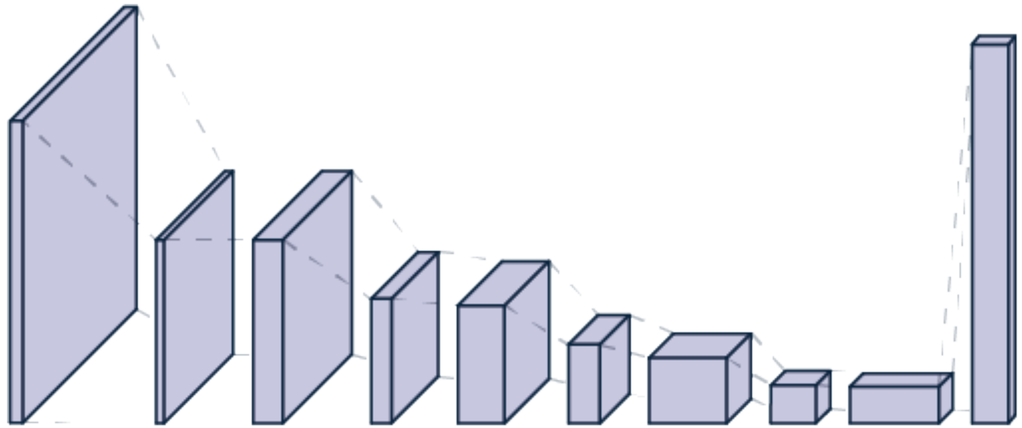
Making data linearly separable



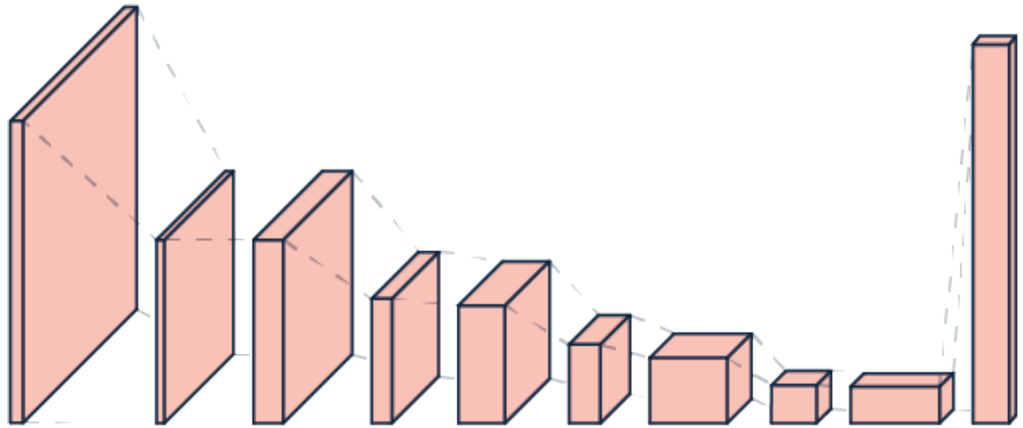
Making data linearly separable

But how?

Which one do you choose?



VS



Train and test representations

Train and test representations



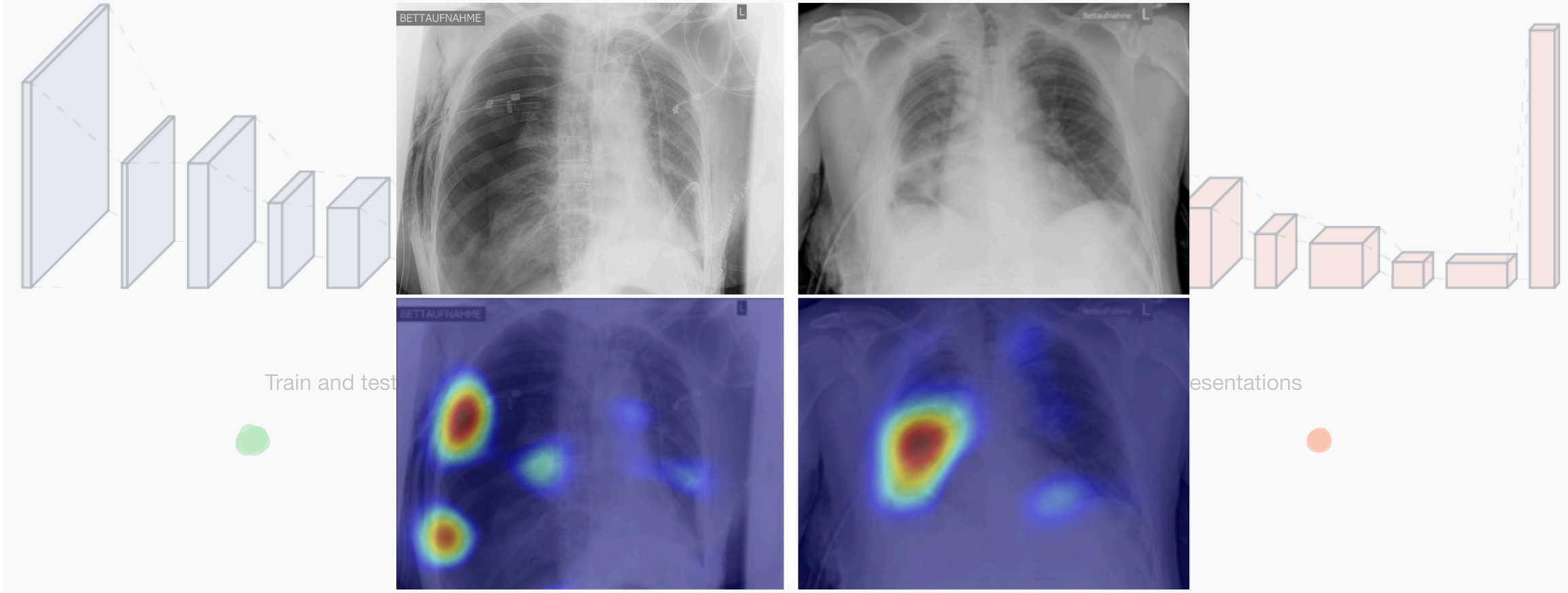


Are these models equivalent?





Which one do you choose?



Train and test esentations

So how do we make sense of what input-level patterns are being represented?

Let’s link functional and representational

Let’s look at some feature maps

Let’s look at some feature maps

(First feature map for a few different samples)

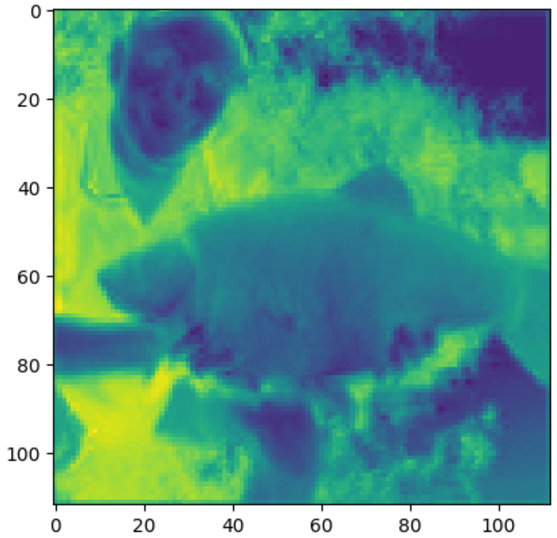
ResNet-18 trained on ImageNet

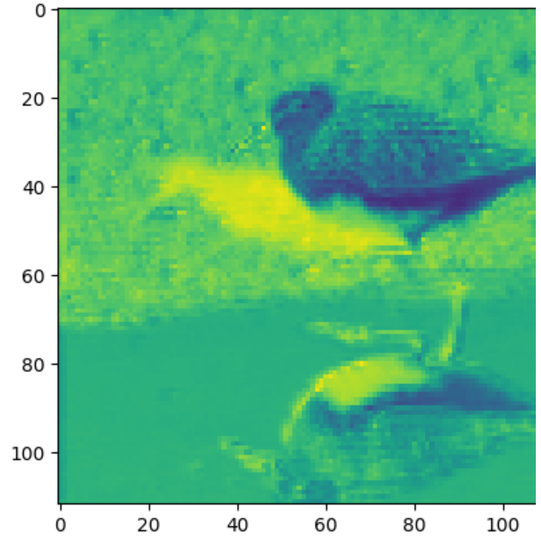
Feature maps after the first convolutional layer



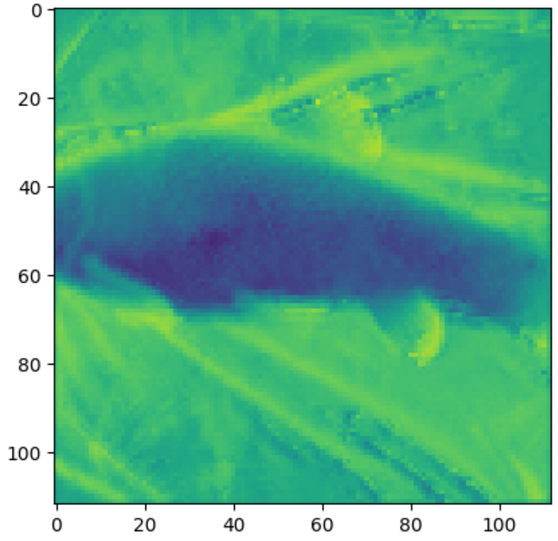
Reference

feature map

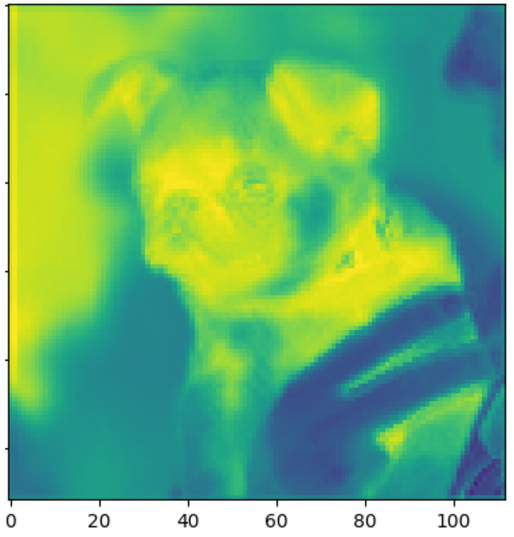




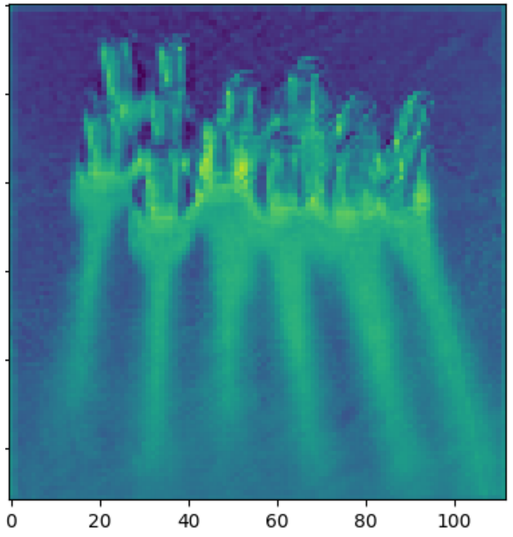
A



B



C



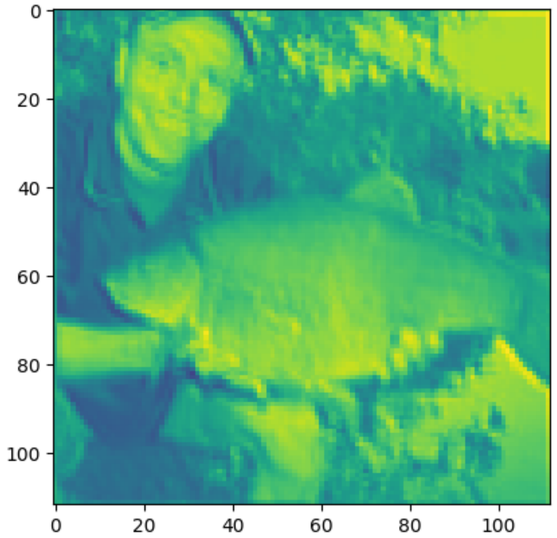
D

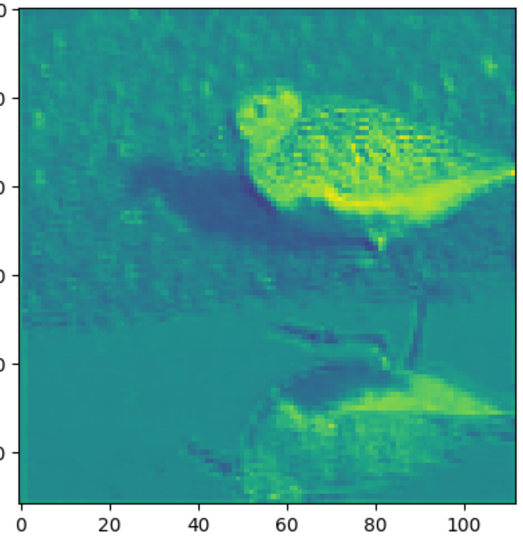
Let’s look at some feature maps

(Second feature map for a few different samples)

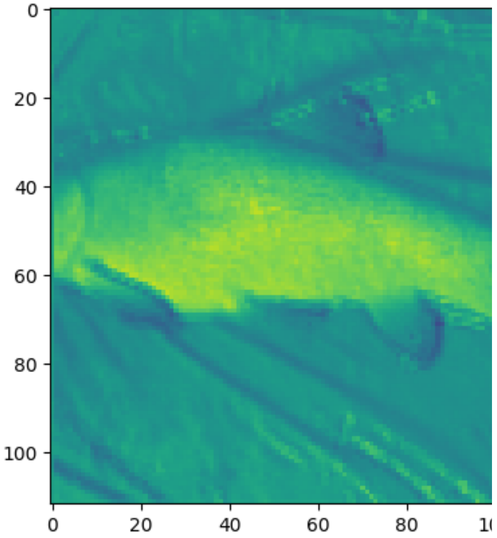
Reference

feature map

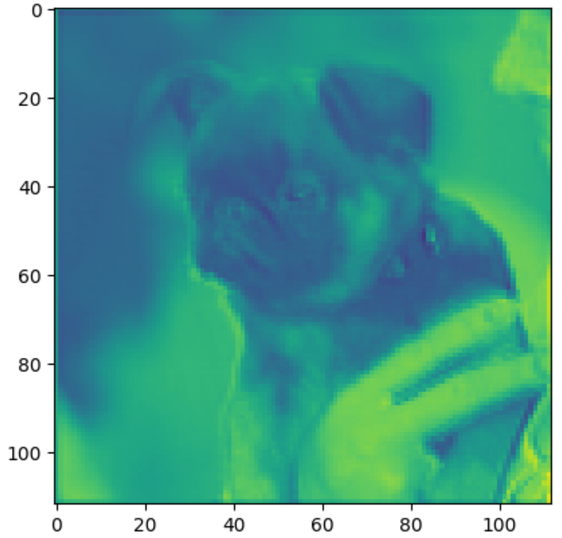




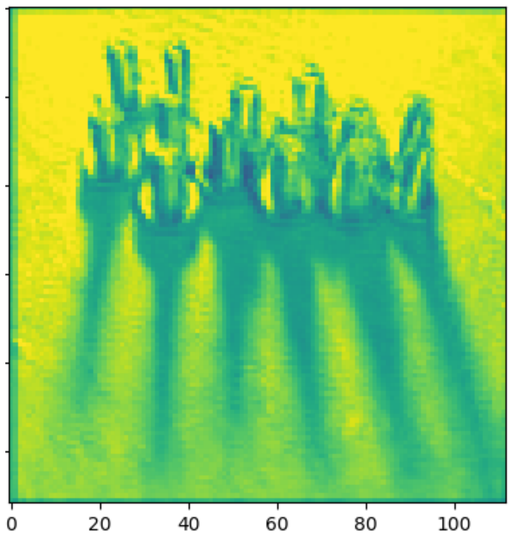
A



B



C

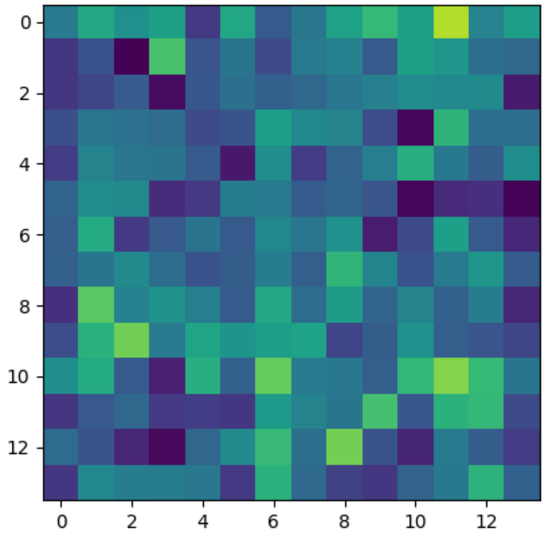


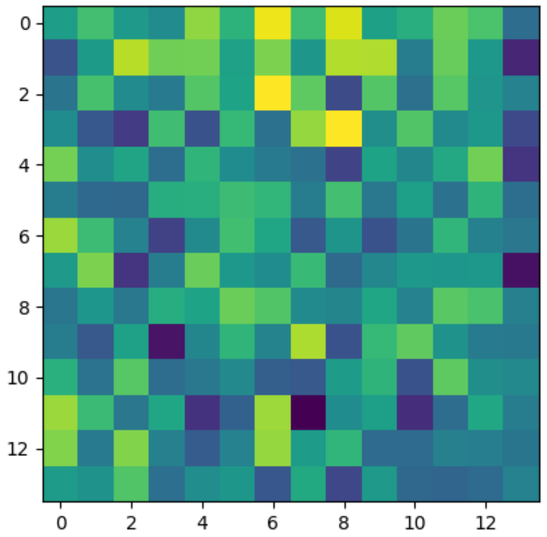
D

Now let’s look at a later point in the network

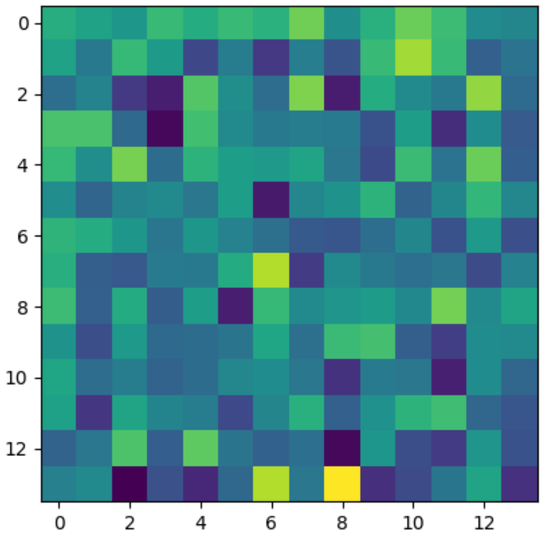
Reference

feature map

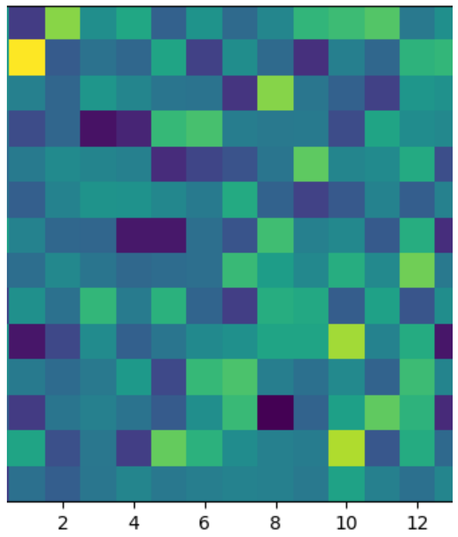




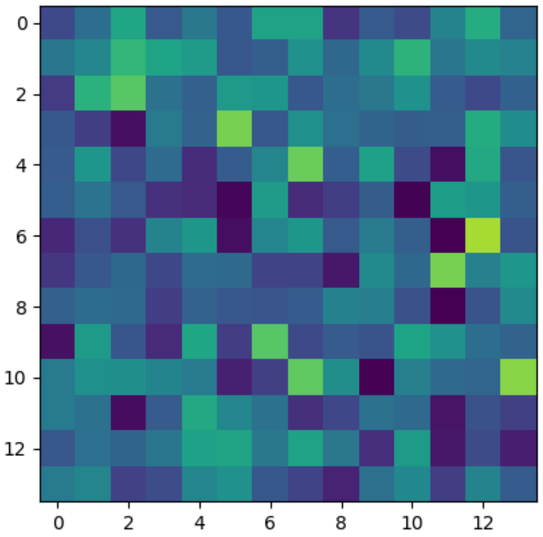
A



B



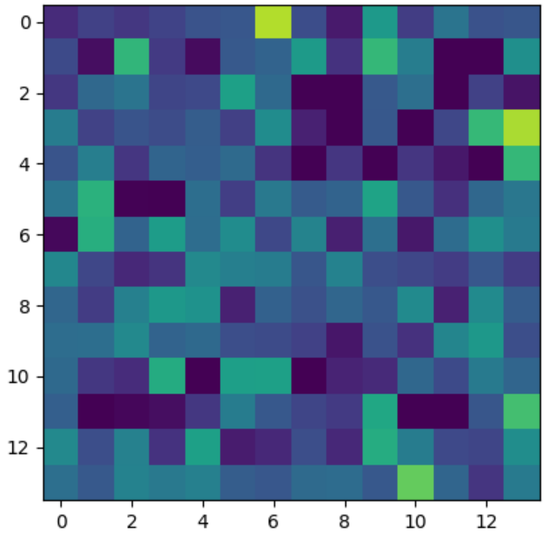
C

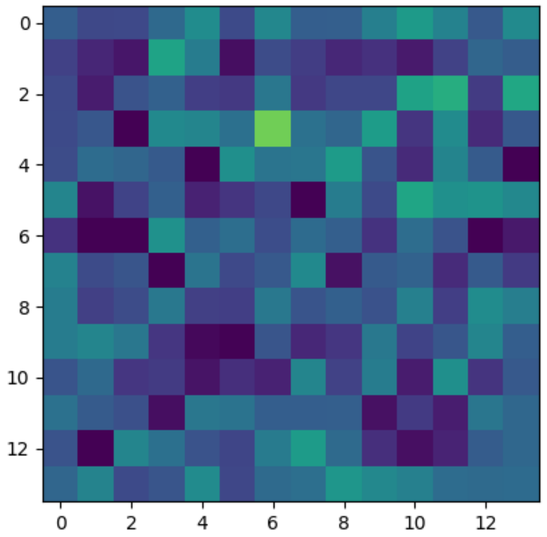


D

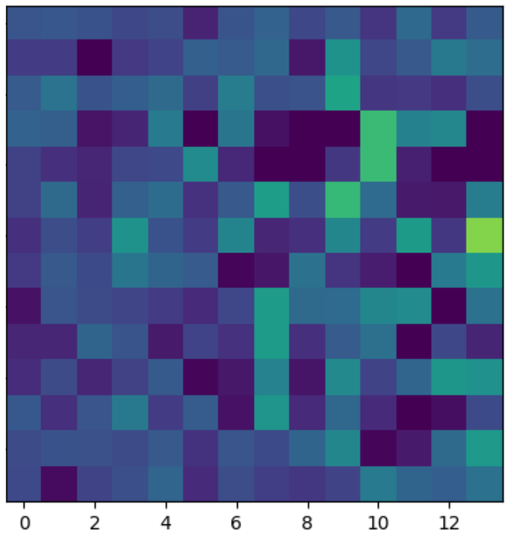
Reference

feature map

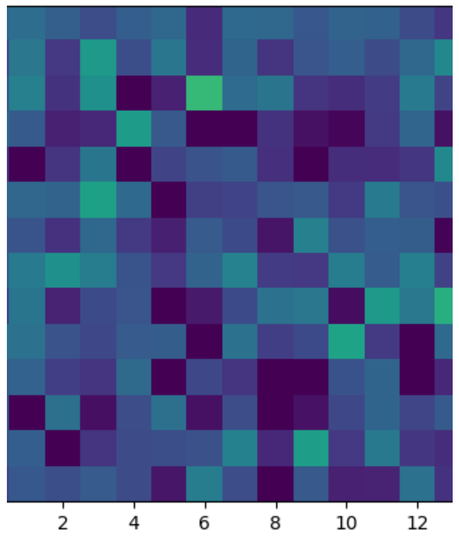




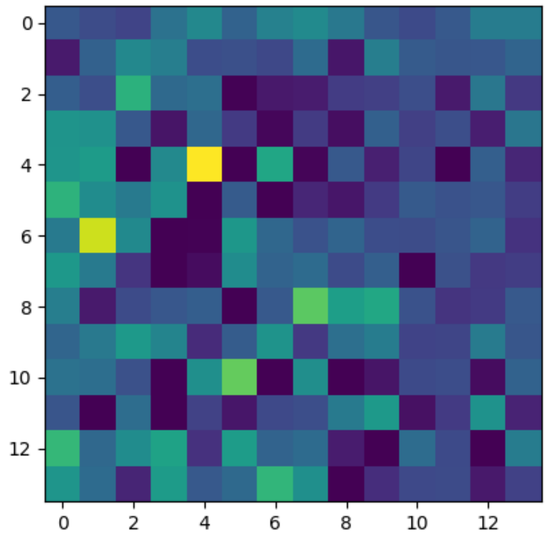
A



B



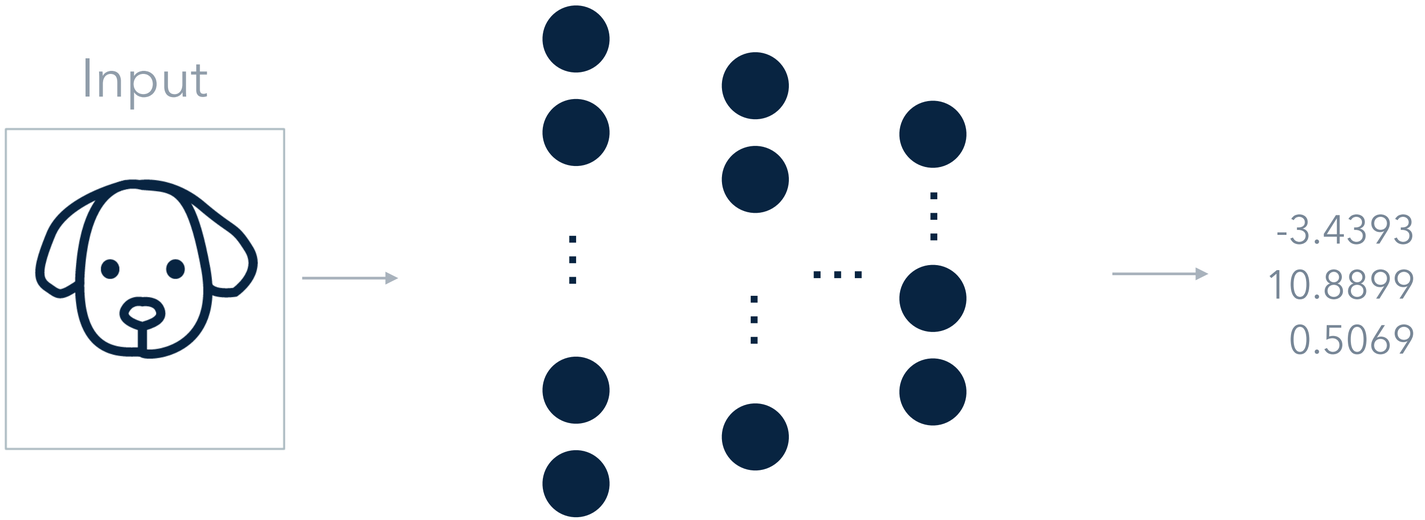
C



D

It’s difficult to make sense of what learned representations actually mean

Visualising “what a network is responding to”



Visualising “what a network is responding to”



Visualising “what a network is responding to”



Visualising “what a network is responding to”



Visualising “what a network is responding to”



Visualisations from: https://distill.pub/2017/feature-visualization/

Visualising “what a network is responding to”



Visualisations from: https://distill.pub/2017/feature-visualization/

Visualising “what a network is responding to”



Visualisations from: https://distill.pub/2017/feature-visualization/

Visualising “what a network is responding to”

How do we choose what we are optimising against?

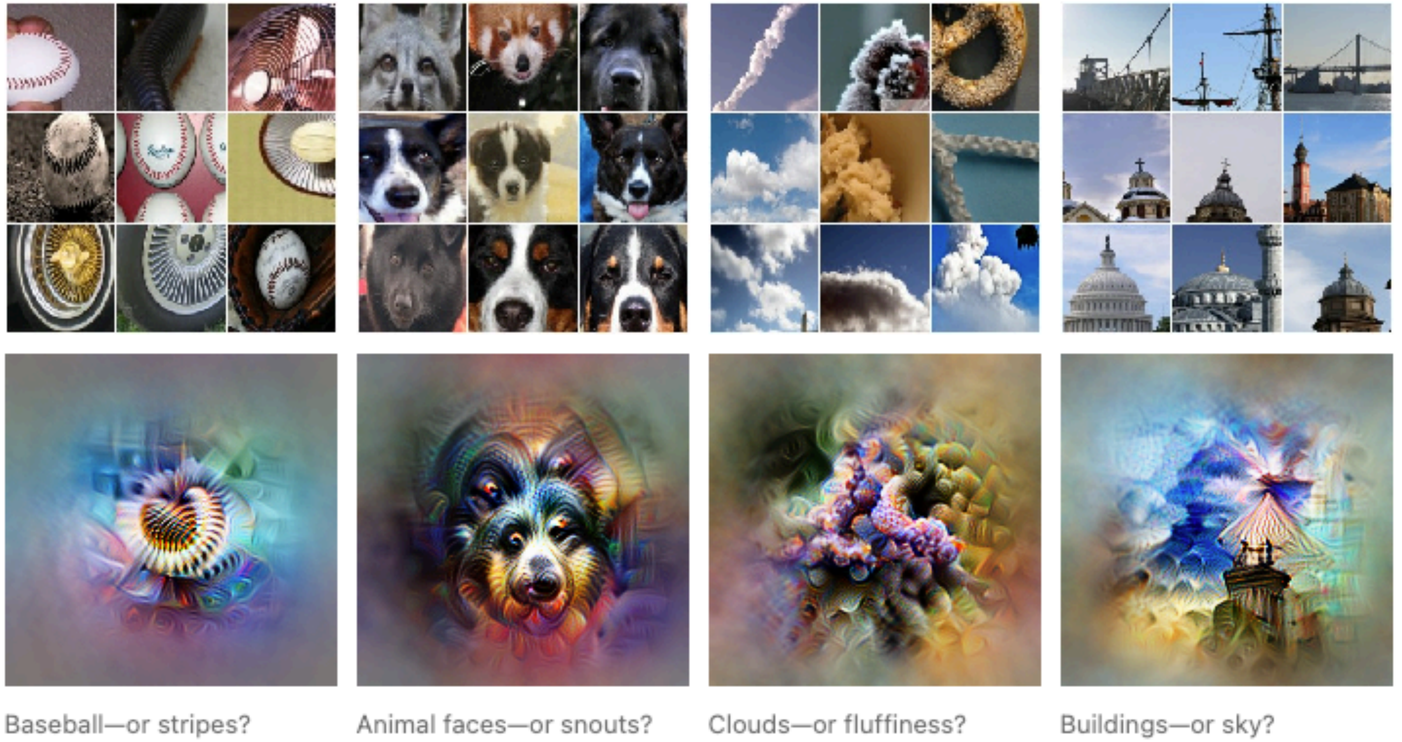
Visualising “what a network is responding to”

How do we choose what we are optimising against?

“If we want to understand individual features, we can search for examples where they have high values — either for a *neuron* at an individual position, or for an entire *channel*.”

Visualisations from: https://distill.pub/2017/feature-visualization/

“**Optimization** isolates the causes of behavior from mere correlations.”



Visualisations from: https://distill.pub/2017/feature-visualization/

Not assessed, but have a look if you are interested in evidence that visualisations can be fooled

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Don’t trust your eyes: on the (un)reliability of feature visualizations

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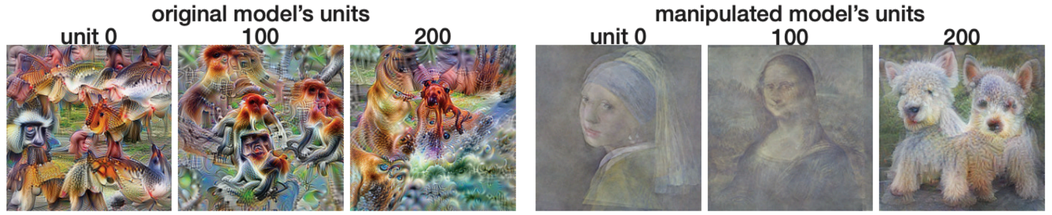


Figure 1: **Arbitrary feature visualizations.** Don’t trust your eyes: Feature visualizations can be arbitrarily manipulated by embedding a fooling circuit in a network, which changes visualizations while maintaining the original network’s ImageNet accuracy. **Left:** Original feature visualizations. **Right:** In a network with a fooling circuit as described in Section 2.1, feature visualizations can be tricked into visualizing arbitrary patterns (e.g., Mona Lisa).

Visualisations from: https://distill.pub/2017/feature-visualization/

Visualising “what a network is responding to”

What is the implicit assumption here?

“If we want to understand individual features, we can search for examples where they have high values—either for a *neuron* at an individual position, or for an entire *channel*.”

Visualisations from: https://distill.pub/2017/feature-visualization/

- Many examples of the “grandmother cell”/ “Jennifer Aniston neuron” (A famous Golden Gate neuron in LLMs)

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- Most representations in neural networks are entangled (not monosemantic)

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- Most representations in neural networks are entangled (not monosemantic)
- This is not problematic just for interpretability, but it impacts our wider ability to make grounded statements about what our models do
- and to create rigorous evaluations

- For example, it’s often useful to know when two models are similar
- We don’t have good tools for measuring model similarity
- Nor for benchmarking similarity tools
- (We will briefly come back to this in the transfer learning seminar next week)